Sachit Mahajan

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SKILLS

Programming: C++, Python, C, Embedded C, MATLAB

Software and Framework: Pytorch, ROS/ROS2, PCL, Eigen, Gazebo, Isaac Sim, RTOS, OpenCV, Ceres, GTSAM, Git, Docker

EXPERIENCE

Amazon.com Services LLC – SDE (I → II → III) , Fauna Robotics *Sunnyvale, CA | Jun 2021 – Present*

- Driving next-generation spatial perception and embodied AI R&D for general-purpose humanoid robots, and building scene representations for 3D spatial, temporal, and semantic reasoning in long-term autonomy.
- Built a 3D Gaussian Splatting based digital twin pipeline that renders robot trajectories in real homes with ground truth poses, dense stereo depth, and semantic labels, scaling data for training and testing.
- Evaluated and fine-tuned learned visual-navigation policies (residual RL, visual navigation transformer) for a humanoid robot in Isaac Sim and Lab, with **1.5-2x** the success of an A*/Nav2 baseline around obstacles.
- Developed and shipped a **SLAM** module for Amazon's consumer robots, deployed to **thousands** of customers.
- Designed and deployed a robust pose graph optimization (RPGO) system to correct odometry drift, increasing stability and map accuracy by over **2.5x** by rejecting outliers. Optimized compute performance under hardware constraints.
- Built SLAM benchmarking pipelines, a long-term autonomy simulation framework, metrics framework and deterministic replay tools for root-cause analysis, improving stability metrics by **2x**.
- Built an AI-driven evaluation framework for autonomy behaviors (LLM-as-a-Judge, statistical testing) with **99.5%** sim-to-real agreement; a golden-set selection method cut evaluation time by **70%** and runs on every code review.

Amazon.com Services LLC – SDE Intern, Consumer Robotics (CoRo) *Remote | May 2020 – Aug 2020*

- Developed planar feature extraction and tracking (walls, floors, ceilings) to improve visual-inertial odometry in low-texture indoor environments, and integrated planar constraints into the SLAM pipeline, reducing relative pose error (RPE).
- Designed a geometric-prior loop closure validation module that reduced false positives in place recognition.

GreyOrange Pte Ltd. – R&D Embedded Engineer, Robotics *Gurgaon, India | Oct 2017 – Jul 2019*

- Developed embedded firmware for warehouse Autonomous Mobile Robots on an RTOS, covering navigation, system diagnostics, obstacle detection, and predictive emergency response modules for safe robot behavior.

EDUCATION

Carnegie Mellon University (CMU) – School of Computer Science – Robotics Institute *Pittsburgh, PA | May 2021*

MS in Robotic Systems Development (MRSD)

Vellore Institute of Technology (VIT) *Vellore, India | July 2013 - May 2017*

B.Tech, Electronics & Communication Engineering

PUBLICATIONS

[1] **POp-GS: Next Best View in 3D-Gaussian Splatting with P-Optimality** (***CVPR 2025***), Joey Wilson, Marcelino Almeida, **Sachit Mahajan**, Martin Labrie, Maani Ghaffari, Omid Alizadeh, Min Sun, Cheng-Hao Kuo, Arnie Sen.

[2] **Modeling Uncertainty in 3D Gaussian Splatting through Continuous Semantic Splatting** (***ICRA 2025***), Joey Wilson, Marcelino Almeida, Min Sun, **Sachit Mahajan**, Maani Ghaffari, Parker Ewen, Omid Alizadeh, Cheng-Hao Kuo, Arnie Sen.

[3] **Design of a Medical Assistant Robot** (***RTEICT 2017***), **Sachit Mahajan**, C.M. Vidhyapathi.

Reviewer for IEEE RA-L, IROS, ICRA, Humanoids, AAAI, CoRL

PROJECTS

[1] **Navigation, Payload Handling and Docking, and Safety Behaviors for an Autonomous Vehicle ( )**: Developed localization, navigation, and precision-docking behaviors for an Autonomous Vehicle operating with modular payloads in geo-fenced environments.

[2] **SALSA - Semantic Assisted Lifelong Structure Aware SLAM ( ):** Learning-augmented lifelong SLAM using semantic landmarks to classify dynamic/repeatable objects. Incorporated high-confidence structural anchors to robustly localize with fewer observations.

[3] **Handling Mirrors in Indoor LiDAR Mapping**: Built a LiDAR-camera pipeline to detect mirrors and correct point-cloud maps, using a novel coarse-to-fine region-growing algorithm for mirror boundary estimation and artifact correction.